

Archaeologist

Genealogy · CLEAR vs TENTATIVE · motion FLAG · exam bite

Genealogy one-liner

Behavior-style human sim authoring || GenSim code-gen → RoboGen full-stack generative simulation
→ infinite synthetic skill stream opposite UMI/DROID real poles.

Big-picture lineage

Human suites → FM writers → RoboGen full-stack → synthetic stream

Human-authored
RLBench / Behavior-100

FM semantic writers
Eureka / GenSim

Generative Sim
paradigm (Xian'23)

RoboGen
propose→learn

→ full-stack scenes+rewards+MP/RL/traj-opt →

|| contrast

OXE / DROID / UMI
REAL data poles

Synthetic skill demos
opposite real poles

Before RoboGen — authoring hunger

Human-authored benchmarks as foil

RLBench
106 tasks

human-authored tabletop

MetaWorld
50

MT10/MT50 suite

ManiSkill2
20

manip diversity limited

Behavior-100
100

household ecology foil
Table 1 direct compare

Behavior-1K
1000

scalability pain RoboGen
cites as hunger

Procedural TAMP
limited

pick-place diversity foil

GenSim concurrent foil

CLEAR related-work + Table 1 · narrower tabletop

GenSim

- LLM writes manip code
- Tabletop pick-place focus
- Small Ravens asset pool
- Self-BLEU 0.378 (worse)
- 70 tasks in Table 1

RoboGen advance

- Full-stack: scene + reward + algo
- Articulated + soft + loco
- Objaverse retrieval + VLM verify
- Self-BLEU 0.284 (best)
- BIT* MP first-class (Fig. 5)

Also in the prior soup (CLEAR citations)

Code-as-Policies · Inner Monologue · VoxPoser · Language-to-Rewards · Eureka · SayCan · Gen2Sim · PartNet-Mobility / Objaverse / Zero-1-to-3

Intellectual move — full-stack generative sim

Not just reward code · scenes + algo select + learn loop

RoboGen: Towards Unleashing Infinite Data for Automated Robot Learning via Generative Simulation

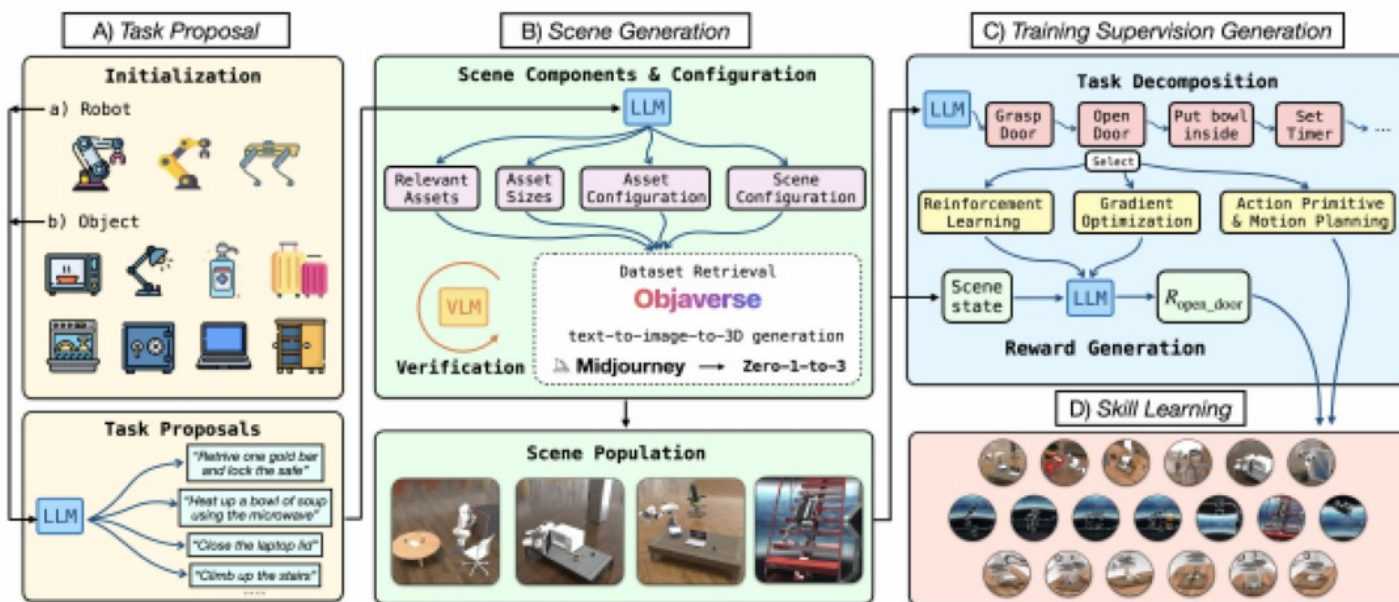


Figure 2: RoboGen consists of the following stages: A) task proposal, B) scene generation, C) training supervision generation, and D) skill learning with generated information.

from its URDF file, and 3) semantic annotations of the links in the object's articulation, e.g., which link corresponds to the door in a sampled microwave. These information are provided by the PartNetMobility dataset. Additionally, we include one example input-output pair in the prompt. We

3.2. Scene Generation

Once a task proposal is obtained, RoboGen then generates a corresponding scene for solving the task by populating the environment with a number of relevant and necessary

CLEAR vs TENTATIVE labels

Archaeologist discipline — do not fabricate

CLEAR

- Instantiates Generative Simulation (Xian'23a)
- Table 1 vs Behavior-100 (+ suites + GenSim)
- Motion planning first-class + Fig. 5
- Long-horizon sequential sub-skills
- Project site + GitHub PyBullet RoboGen
- Contrast to UMI/DROID/OXE (session genealogy)
- GenSim concurrent narrower foil

TENTATIVE

- Packaged skill-library API product
- Full Genesis public soft-body timing
- Official BEHAVIOR'26 data pipeline
- Frozen public demo-count / hours
- Unique descent of later LLM-reward systems
- Zero human involvement claim
- Sim-to-real % solved

Opposite real poles — session genealogy

CLEAR contrast · not a RoboGen empirical claim

UMI

Handheld human demos

Wild visual + action
~\$371 gripper

DROID

Franka in buildings

564 scenes / 76k
protocol diversity

OXE

Multi-lab agglomeration

22 robots · community
breadth soup

RoboGen

Generative sim factory

Endless skill demos
where teleop too costly

Motion / skill-library FLAG — PRIMARY

Not optional garnish · CLEAR evidence

CLEAR Algo menu includes BIT* MP primitives (grasp/approach/release)

CLEAR Long-horizon = sequential sub-skills · N=8 best-state handoff

CLEAR Fig. 5: MP critical — RL-only collapses on 12 articulated

CLEAR Avg structure $\approx$ skill atoms: 1.63 MP + 1.5 RL · 3.13 total

TENTATIVE “Skill library” as packaged reusable API — paper emits demos/skills

CLEAR Genesis multi-material: internal; public PyBullet for rigid+loco

Evidence pair — Fig. 5 + Fig. 6

Collapse without MP · hybrid substep mix is the product

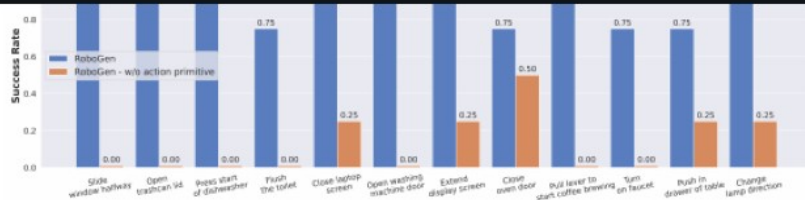


Figure 5: Among 12 articulated object manipulation tasks, the success rate decreases drastically if only RL is used for skill learning.

tion tasks, meanwhile leverage more diverse assets retrieved from open-world databases such as Objaverse, resulting in much higher diversity in both task semantics and scene

judge whether the joint angle value 0 corresponds to the door being opened or closed. 3) failure to find matched assets for tasks that require extremely precise spatial relationships,

RoboGen: Towards Unleashing Infinite Data for Automated Robot Learning via Generative Simulation

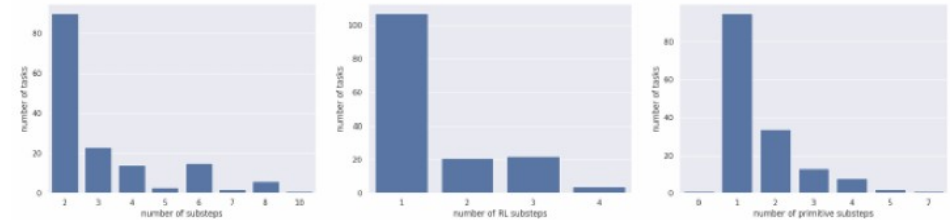


Figure 6: Left: The distribution of number of substeps for the generated rigid and articulated object manipulation tasks in Table 2. The average number of substeps is 3.13. Middle: The distribution of number of substeps that need to be solved using RL for the generated tasks. The average number of RL substeps is 1.5. Right: The distribution of number of substeps that need to be solved using motion planning based primitives for the generated tasks. The average number of such kind of substeps is 1.63. Regarding duration for solving the task: if the task's subgoals can all be solved via planning, typically each task can be solved within 10 minutes. If certain subgoals require RL to solve, it usually takes around 2-3 hours for each RL-necessary step, and the total duration thus depends on both the number and nature of the subtasks. Taking these into account, a task typically takes 4-5 hours on average. This is done using 8 threads of a CPU running at 2.5Ghz, meaning that each single node in a cluster with a 32-core (64 threads) CPU could run 8 jobs in parallel at the same time.

Fig. 5 RL-only collapse

Exam bite (CLEAR)

RoboGen's long-horizon competence is decomp + choose MP or RL or traj-opt per sub-skill — not a single end-to-end LLM policy.

Fig. 6 ~1.5 RL + ~1.63 MP

Fig. 3 — long-horizon qualitative

Decomp + reward validity on 4 example skills

RoboGen: Towards Unleashing Infinite Data for Automated Robot Learning via Generative Simulation

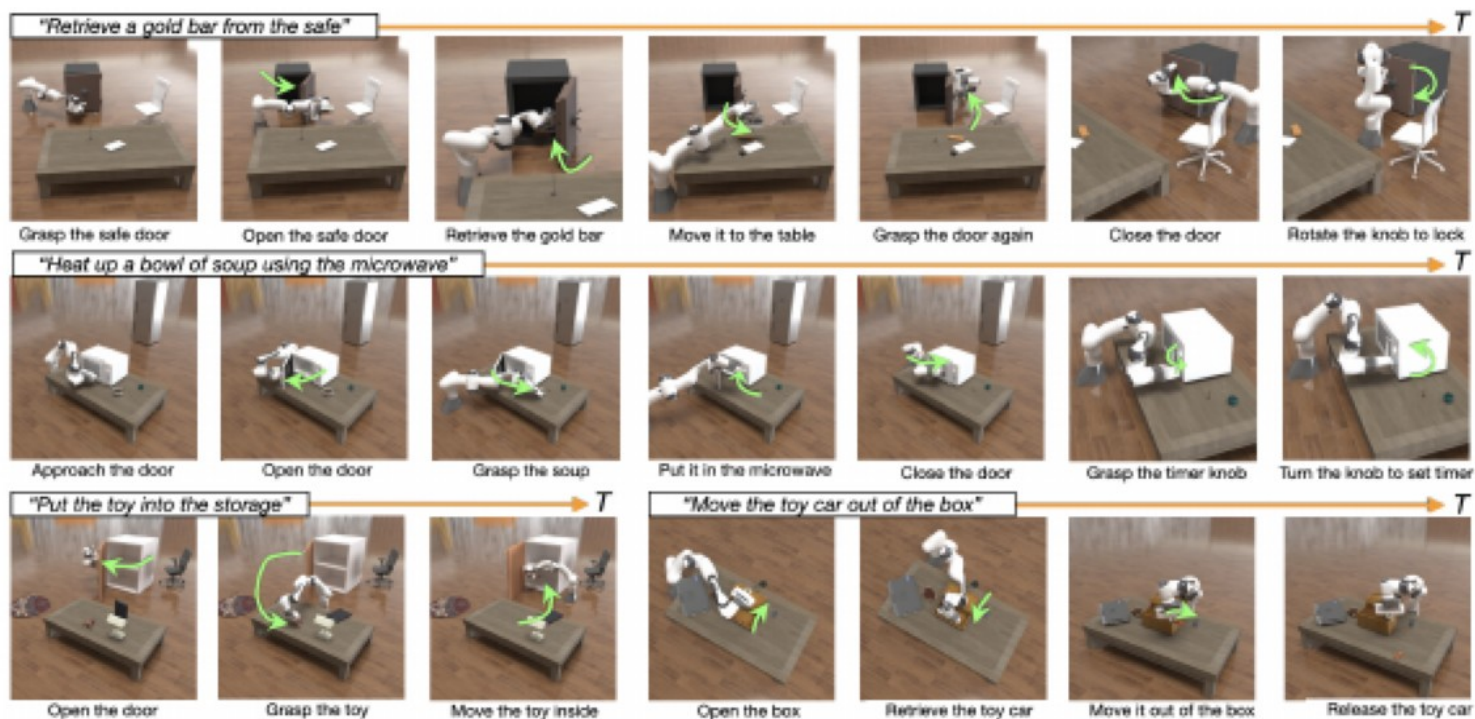


Figure 3: Snapshots of the learned skills on 4 example long-horizon tasks.

RoboGen Behavior-100 RLbench MetaWorld Maniskill2 GenSim

Descendants / ecosystem

RoboGen PyBullet repo · CLEAR site/GitHub · TENTATIVE influence claims · robogen-ai.github.io

Genesis-Embodied-AI/RoboGen ·
rigid+loco public

Project gallery · CLEAR web

Genesis engine

Announced; soft-body release timing
TENTATIVE

Downstream consumers

DP / ACT / π_0 / OpenPI as BC/FT of
demos

BEHAVIOR hunger

Design prior for automating
task+scene+demo minting

Do not over-claim

Unique descent of later systems
without citation checks

Exam bite — Archaeologist

Memorize this card

RoboGen = GPT-4 proposes tasks/scenes/rewards; physics + SAC/BIT*/Adam learn skills; diversity beats Behavior-100 on Table 1 (Self-BLEU 0.284); ~77% skill-run success (0.774/69); MP ablation matters (Fig. 5). Lineage: Behavior human authoring || GenSim → RoboGen full-stack → synthetic stream opposite UMI/DROID/OXE.

Numbers tattoo — pin these

Every deck · Ablation Bot MISSING (paper extract)

0.284

Self-BLEU (best)

0.165

SentBert ↓

0.193

Scene ViT ↓

0.762

Scene CLIP ↓

0.774

success / 69

19 / 155

validity fails

3.13

avg substeps

Fig. 5

RL-only collapses

0.745

manip SR (n=50)

~0.886

soft SR (n=7)

~0.833

loco SR (n=12)

~1.5+1.63

RL + MP avg

Archaeologist — close

Takeaways + exam bite

1. Genealogy: Behavior authoring || GenSim → RoboGen full-stack → synthetic opposite real poles.
2. Label CLEAR vs TENTATIVE — packaged skill API and BEHAVIOR'26 official are TENTATIVE.
3. Motion FLAG is PRIMARY: Fig. 5 + 3.13 substep hybrid structure.
4. Exam: decomp + choose MP/RL/traj-opt — not E2E LLM policy.

EXAM BITE

Behavior || GenSim → RoboGen → synthetic stream $\perp$ UMI/DROID/OXE.